

Sara Klaasen

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Education

PhD	ETH Zurich , Geophysics • ETH medal	Zurich, Switzerland 2020 – 2025
Msc	ETH Zurich , Geophysics	Zurich, Switzerland 2018 – 2020
Bsc	UCR , Liberal Arts and Sciences • Magna cum Laude • Major in Earth sciences, biology, mathematics and physics	Middelburg, Netherlands 2014 – 2017

Experience

Research School of Earth Sciences, Australian National University , Postdoctoral researcher Researching DAS projects in New Zealand. • Imaging Taupo Volcano.	Canberra, Australia 2026 – present 1 year
ETH Zurich , Postdoctoral researcher	Zurich, Switzerland 2025 – 2026 1 year
SRO , Admin • Online administration	Haarlem, Netherlands 2017 – 2017 1 year
PZC , Online editor • Online content management of the provincial newspaper • Writing and editing news articles	Vlissingen, Netherlands 2015 – 2017 2 years

Awards

ETH Medal The ETH Medal is given to outstanding doctoral theses. ETH Zurich	2026
OSPA Outstanding student presentation award Seismological Society of America Photonics Meeting	2024

Grants

SNF postdoc.mobility grant A grant to fund a 2-year postdoc position. Swiss National Science Foundation	2025
VSBFonds grant A grant to support a Masters programme abroad. VSBFonds	2018

Books

- Fiber optic sensing for volcano monitoring and imaging volcanic processes** 2025
Book chapter in Modern Volcano Monitoring
Philippe Jousset, Gilda Currenti, Shane Murphy, Eva Eibl, Corentin Caudron, Lise Retailleau, Christopher Wollin, Sara Klaasen, Takeshi Nishimura, Andreas Fichtner, Zack Spica, Arnaud Lemarchand, Charlotte Krawczyk
doi.org/10.1007/978-3-031-86841-2_17 (Springer)

Publications

- Submarine seismicity monitoring with distributed acoustic sensing near Santorini and Kolumbo Volcano** 2026
Sara Klaasen, Thomas Hudson, Sanchit Sachdeva, Paraskevi Nomikou, Andreas Fichtner
doi.org/10.26443/seismica.v5i1.1845 (Seismica)
- Seismic constraints on glacier density** 2025
Ariane Lanteri, Scott Keating, Lars Gebraad, Sara Klaasen, Marta Pienkowska-Cote, Olaf Eisen, Andrea Zunino, Kristín Jónsdóttir, Coen Hofstede, Dimitri Zigone, Andreas Fichtner
doi.org/10.1038/s41598-025-26440-z (Scientific Reports)
- The Seismogenic Thickness of Venus** 2025
Julia Maia, Ana-Catalina Plesă, Iris van Zelst, Richard Ghail, Anna Gülcher, Mark Panning, Sven Peter Näsholm, Barbara de Toffoli, Anna Horleston, Krystyna Smolinski, Sara Klaasen, Robert Herrick, Raphael Garcia
doi.org/10.1038/s41598-025-26440-z (Journal of Geophysical Research: Planets)
- Towards a widely applicable earthquake detection algorithm for fibreoptic and hybrid fibreoptic-seismometer networks** 2025
Thomas Hudson, Sara Klaasen, Olivier Fontaine, Conor Bacon, Kristin Jonsdottir, Andreas Fichtner
doi.org/10.1093/gji/ggae459 (Geophysical Journal International)
- An illustrated guide to: Distributed and integrated fibre-optic sensing in seismology** 2024
Andreas Fichtner, Fabian Walter, Patrick Paitz, Sara Klaasen, Daniel Bowden, Sebastian Noe, Nils Müller, Dominik Husmann, Jacques Morel
dx.doi.org/10.1016/j.eq.s.2024.09.006 (Earthquake Sciences)
- Seismic wave detectability on Venus using ground deformation sensors, infrasound sensors on balloons and airglow imagers** 2024
Raphael Garcia, Iris van Zelst, Taichi Kawamura, Sven Peter Näsholm, Anna Horleston, Sara Klaasen, Maxence Lefevre, Celine Marie Solberg, Krystyna Smolinski, Ana-Catalina Plesă, Quentin Brissaud, Julia Maia, Simon Stähler, Philippe Lognonné, Mark Panning, Anna Gülcher, Richard Ghail, Barbara de Toffoli
doi.org/10.1029/2024EA003670 (Earth and Space Science)
- Modelling uncertainty in P-wave arrival-times retrieved from DAS data: case-studies from 15 fibre optic cables** 2024
Emanuele Bozzi, Nicola Piana Agostinetti, Andreas Fichtner, Sara Klaasen, Arantza Ugalde, Biondo Biondi, S Yuan, Torsten Dahm, Marius Isken, Patrick Paitz, Fabian Walter, Alain Baird, Carlos Becerril, Takeshi Nishimura, Junzhu Shen, Tieyuan Zhu, Gilberto Saccorotti
doi.org/10.1093/gji/ggae364 (Geophysical Journal International)

Complex spatial distribution of onset amplitude and waveform correlation: case studies from different DAS experiments Emanuele Bozzi, Gilberto Saccorotti, Nicola Piana Agostinetti, Carlos Becerril, Andreas Fichtner, Sara Klaasen, Takeshi Nishimura, Junzhu Shen, Fabian Walter, Tieyuan Zhu doi.org/10.4430/bgo00458 (Bulletin of geophysics and oceanography)	2024
Challenges in submarine fiber-optic earthquake monitoring Jonas Igel, Sara Klaasen, Sebastian Noe, Paraskevi Nomikou, Konstantinos Korantzalos, Andreas Fichtner doi.org/10.1029/2024JB029556 (Journal of Geophysical Research: Solid Earth)	2024
Subglacial volcano monitoring with fibre-optic sensing: Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristín Jónsdóttir, Lars Gebraad, Patrick Paitz, Andreas Fichtner doi.org/10.30909/vol.06.02.301311 (Volcanica)	2023
Fiber-optic observation of volcanic tremor through floating ice sheet resonance Andreas Fichtner, Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Patrick Paitz, Kristin Jonsdottir doi.org/10.1785/0320220010 (The Seismic Record)	2022
Distributed acoustic sensing in volcano-glacial environments — Mount Meager, British Columbia Sara Klaasen, Patrick Paitz, Nadja Lindner, Jan Dettmer, Andreas Fichtner doi.org/10.1029/2021JB022358 (Journal of Geophysical Research: Solid Earth)	2021
Detecting Seismic Events with Computer Vision: Applications for Fiber-Optic Sensing Solvi Thrastarson, Robert Torfason, Sara Klaasen, Patrick Paitz, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner doi.org/10.1002/essoar.10509693.1 (ESS Open Archive)	2021
Conference contributions (selected)	
Full-Waveform Inversion for source and structure of DAS data on a subglacial volcano: Grímsvötn, Iceland invited talk Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner (IASPEI-IAGA Scientific Assembly)	2025
The potential of distributed acoustic sensing for volcano monitoring and imaging Keynote talk Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Lars Gebraad, Thomas Hudson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jan Dettmer, Paraskevi Nomikou, Andreas Fichtner (IAVCEI General Meeting)	2025
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (IAVCEI General Meeting)	2025
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (IAVCEI General Meeting)	2025

Full-Waveform Inversion of DAS data on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Andreas Fichtner (IAVCEI General Meeting)	2025
Submarine volcano monitoring with Distributed Acoustic Sensing at Kolumbo, Greece Sara Klaasen, Thomas Hudson, Paraskevi Nomikou, Andreas Fichtner (SSA General Meeting)	2025
Fibre-Optic Seismology on Remote, Subglacial, Submarine and Actively-Erupting Volcanoes invited talk, outstanding student presentation award Sara Klaasen, Kristin Jonsdottir, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Lars Gebraad, Andreas Fichtner (SSA photonic seismology)	2024
Moment Tensor inversion with Full-Waveform Inversion and Distributed Acoustic Sensing on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU Copernicus Meetings)	2024
Submarine volcano monitoring with fiber-optic cables at Santorini Sara Klaasen, Paraskevi Nomikou, Konstantinos Karantzas, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	2023
Going towards Full Waveform Inversion (FWI) with Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Sebastian Noe, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Jonas Igel, Andreas Fichtner (AGU Annual Meeting)	2023
Volcano monitoring and modelling using Distributed Acoustic Sensing (DAS) on a subglacial volcano: Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Sebastian Noe, Andreas Fichtner (IUGG General Assembly)	2023
Monitoring glacially covered volcanoes with Fibre-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (IAVCEI General Meeting)	2023
Volcanic Event Characterization with Fiber-Optic Sensing Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (AGU Annual Meeting)	2022
Fibre-Optic Sensing for Volcano Monitoring on Grímsvötn, Iceland Sara Klaasen, Solvi Thrastarson, Yesim Cubuk-Sabuncu, Kristin Jonsdottir, Lars Gebraad, Andreas Fichtner (EGU General Assembly)	2022

Distributed Acoustic Sensing on Volcanoes: Grímsvötn and Fagradalsfjall, Iceland Sara Klaasen, Solvi Thrastarson, Kristin Jonsdottir, Yesim Cubuk-Sabuncu, Patrick Paitz, Andreas Fichtner (AGU Annual Meeting)	2021
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner (SSA General Meeting)	2021
Combining Distributed Acoustic Sensing and Beamforming in a Volcanic Environment on Mount Meager, British Columbia Sara Klaasen, Patrick Paitz, Jan Dettmer, Andreas Fichtner (EGU General Assembly)	2021
Distributed Acoustic Sensing for the Exploration of the Mount Meager Volcanic Complex, British Columbia, Canada Sara Klaasen, Andreas Fichtner, Patrick Paitz, Jan Dettmer (AGU Annual Meeting)	2020

Talks (invited)

Frontiers in Fiber-Optics Sensing in Seismology IASPEI-IAGA, Lisbon (Invited speaker)	2025
ECR Keynote IAVCEI, Geneva (Invited keynote speaker)	2025
Seminar Stanford University, San Francisco (Seminar speaker)	2025
Seminar University of Washington, online (Seminar speaker)	2024
An innovative photonic vision of volcanoes and geothermal systems SSA Photonic Seismology, Vancouver (Invited speaker)	2024
Colloquium CSIRO, Perth (Seminar speaker)	2023
Geophysics Weekly Meeting Australian National University, Canberra (Seminar speaker)	2023
Seminar University of Canterbury, Christchurch (Seminar speaker)	2023

Service

Convening conference sessions

Reviewing

Languages

Dutch

native

English

C2

German

B2

French

B1